Figure 1: The Ionic Framework (*ionicframework.com*)

Figure 2: Technologies underlying the Ionic Framework (*cabotsolutions.com*)

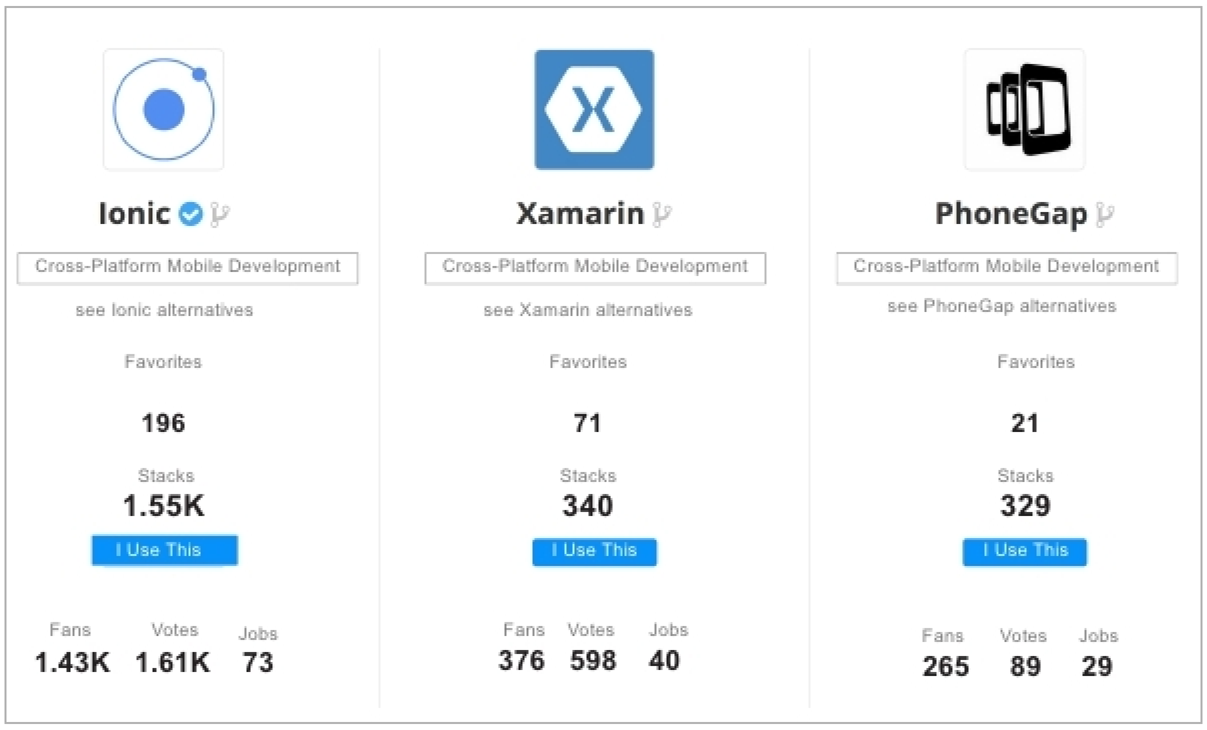

Figure 3: Comparison of hybrid application development frameworks (*stackshare.io*)

Ionic Framework, demonstrating the potential this has to offer for future developers.

Features of the Ionic Framework

The Ionic Framework boasts of a wide range of features that not only offer useful functionality but also address some issues that are specific to using HTML, CSS and JS to mimic native functionality.

For instance, it does away with the dreaded *tap delay*, the reason why most of us experience the seemingly slow response to a tap on an HTML5 application. Besides that, it introduces faster DOM manipulation and menu support out-of-the-box, unlike many other competitors in its category. Finally, and most importantly, the Ionic Framework offers a set of reusable components that permit much faster development than would have been possible had there been a need to create components from scratch. However, these components do not preclude the performance constraints that most applications undoubtedly present with reusable components.

Collection-repeat is one example of how the Ionic Framework provides workarounds to modify conventional Web development practices in order

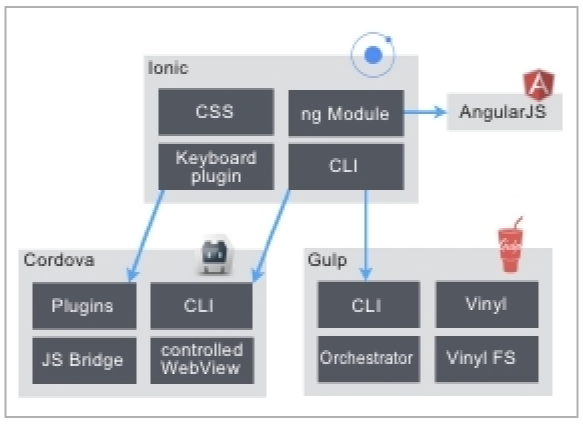

Figure 4: The Ionic Framework architecture (*codecen*

```
<ion-content>
  <ion-item collection-repeat="item in items">
    {{item}}
  </ion-item>
</ion-content>
```

Figure 5: Simple implementation of 'collection-repeat'

to provide a seamless experience for users of the mobile app. It provides a scroll view of hundreds and thousands of items without the performance hits that would be expected for displaying a list on such a scale.

Since the Ionic Framework is based on Apache Cordova, the plugins form the core idea of prebuilt functionality that can compete with native speeds. Cordova was the project that was originally intended to provide a base for writing mobile applications in HTML, CSS and JS. By building on top of this, Ionic offers an extension to its functionality without the need for an additional learning curve, even for experienced users. This makes it a highly lucrative option, especially when starting off, without the need or the bandwidth to support native application development. The Ionic Framework is a project under active development as it constantly needs to stay updated in order to support the latest version of AngularJS, among other components. It is currently on its third version, and has introduced a host of new features such as 'lazy loading'.

It is important to mention that, at the end of the day, Ionic is still just a hybrid application framework and will be hard-pressed to match up to the overall performance and latency of a native application. Quicker development with less effort, however, is what pushes most teams to choose the Ionic Framework. END

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